

Measuring Macroeconomic and Financial Uncertainty: Impact for the Mexican Economic Growth Expectation

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1 Introduction

Central banks when making monetary policy must account for various risks including real economic factors and financial risks. When doing a balance of risks they weigh both downside risks –production, unemployment– and upside risks –inflation. In addition, financial risk are usually associated with high volatility, that can be seen as an endogenous response to the business cycle or as an exogenous impulse to it (Ludvigson et al., 2021).

Measuring uncertainty arises as a problem relevant for decision makers. Three main visions to uncertainty stand out in the literature: the Economic Policy Uncertainty Baker et al. (2016), which tries to answer how uncertain is the policy environment that households and firms face; the VIX index used in finance addresses the question of how much financial markets expect asset prices to fluctuate; while, another view is the one from Jurado et al. (2015) (JLN - hencefort) which is concerned about how unpredictable is the macroeconomy, and Ludvigson et al. (2021) (LMN - henceforth) about what are the roles of macroeconomic and financial uncertainty.

The problem for the forecaster is then extended from the usual GDP growth point estimate to the broader task of estimating lower, middle and higher quantiles. These additional estimates can be later used for assessing the conditional mean of GDP growth on increased volatility of economic and financial conditions. The question at hand is thus: *what is the worst economic growth outcome what we should expect with given probability?* This has been

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tackled in the relevant study by Adrian et al. (2019). They develop a framework –Growth-at-Risk (GaR)– that allow to obtain the empirical quantile distribution based on preliminary lower, middle and upper quantiles.

In this research I focus on the estimation of the empirical quantile distribution of IGAE growth —proxy for monthly GDP— conditional on uncertainty measures for the Mexican economy. The question is relevant because, unlike the US FOMC, the Mexican central bank does not have the dual mandate of keeping low inflation and achieving potential GDP growth. Banco de Mexico (Banxico) has the only mandate of keeping inflation low; thus, its assessment of risks is not the same as in the US economy.

IGAE is a monthly indicator built with most of the data used to compute quarterly GDP and has become relevant in the analysis of short-term economic situation (Elizondo, 2019). In Section 3.1, I present this variable and conduct unit-root tests, test for endogenous structural breaks, and decompose the series into cycle + permanent components.

To construct the indices described before, in Section 3.2 I build on the data of Corona et al. (2022). In particular, I rely on the panel that considers the most relevant information for the Mexican economy, which has two key characteristics: contains only public information, which allows reproducibility of results; it contains information that is timely available, as it is designed to generate nowcasts in the first 15 calendar days after the month closes. In Section 4.1 I employ the same methodology of Corona et al. (2022) to estimate a Dynamic Factors Model (DFM). This methodology is the one currently used by the Mexican national statistics agency, INEGI, to produce timely estimates on a regular basis¹.

With DFM in hand, in Section 4.2 I proceed to employ the JLN methodology to estimate uncertainty indices, which are later employed for the remaining of the paper. As a natural continuation of the JLN methodology, in Section 4.3 I proceed to estimate a SVAR to test the ordering of uncertainty to real economic activity and viceversa, followed by an analysis for their relevance into forecasting and impulse-response function. As a robustness check, in Section 4.4 IRF are estimated using local projections following Jordà (2005).

The Section 4.5 then estimates the yearly growth rate of IGAE conditional on the uncertainty indices constructed in previous sections and the current economic conditions. Finally, the Section 4.6 presents an analysis of real economic activity and labor markets, along with the uncertainty indices measured before.

The paper conclusions and directions for future research are presented in Section 5.

¹<https://www.inegi.org.mx/investigacion/ioae/>

2 Summary of Methodologies

Dynamic Factors Model

Let $\mathbf{x}_t = (x_{1t}, \dots, x_{Nt})'$ be an $N \times 1$ vector of stationary macroeconomic and financial series observed at $t = 1, \dots, T$. The DFM decomposes each variable into a common and an idiosyncratic component:

$$x_{it} = \boldsymbol{\lambda}_i' \mathbf{F}_t + e_{it}, \quad i = 1, \dots, N, \quad (1)$$

where $\mathbf{F}_t = (F_{1t}, \dots, F_{rt})'$ is an $r \times 1$ vector of latent common factors, $\boldsymbol{\lambda}_i$ is the corresponding vector of factor loadings, and e_{it} is the idiosyncratic disturbance, assumed to be weakly correlated across i and uncorrelated with $\mathbf{F}_t$. The number of factors r is selected using the Bai & Ng (2002) information criteria:

$$\text{IC}_p(k) = \ln V(k, \hat{\mathbf{F}}^k) + k \cdot g(N, T), \quad (2)$$

where $V(k, \hat{\mathbf{F}}^k)$ is the average residual variance under k factors and $g(N, T)$ is a penalty function that increases with the panel dimensions. Factors and loadings are estimated by the two-step principal-components method of Doz et al. (2012).

Measuring uncertainty

Following Jurado et al. (2015), uncertainty for series j at horizon h is defined as the conditional volatility of its unforeseeable component. We obtain

$$e_{it} = x_{it} - \mathbf{F}_t' \boldsymbol{\lambda}_i e_{it}.$$

$$\mathcal{U}_{jt}^y(h) = \sqrt{\mathbb{E} \left[(y_{j,t+h} - \hat{\mathbb{E}}[y_{j,t+h} \mid \mathcal{I}_t])^2 \mid \mathcal{I}_t \right]}, \quad (3)$$

where $\hat{\mathbb{E}}[y_{j,t+h} \mid \mathcal{I}_t]$ is the best linear forecast of $y_{j,t+h}$ using information $\mathcal{I}_t = \{\mathbf{F}_s, \mathbf{x}_s\}_{s \leq t}$.

Then, estimate a GARCH(1,1) model. For each variable i , fit: $e_{it} = \sigma_{it} \eta_{it}$, $\eta_{it} \sim iid(0, 1)$

$$\sigma_{it}^2 = \omega_i + \alpha_i e_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2$$

with $\omega_i > 0$, $\alpha_i, \beta_i \geq 0$, $\alpha_i + \beta_i < 1$.

where σ_{it} is the conditional standard deviation of the unforeseeable component, what JLN call the measure of uncertainty.

Next, aggregate each of the volatility estimates into MU and FU

$$MU_t = \frac{1}{N_M} \sum_{i \in M} \sigma_{it}, \quad FU_t = \frac{1}{N_F} \sum_{i \in F} \sigma_{it}$$

Both indexes are standardized to zero mean and unit variance.

Structural VAR

Let $\mathbf{X}_t = (MU_t, IGAE_t, FU_t)'$. The reduced-form VAR(p) is:

$$\mathbf{X}_t = \mathbf{c} + \sum_{j=1}^p \mathbf{A}_j \mathbf{X}_{t-j} + \mathbf{B} \mathbf{d}_t + \mathbf{u}_t, \quad \mathbf{u}_t \sim (0, \Sigma_u), \quad (4)$$

where $\mathbf{d}_t$ is a vector of impulse dummies absorbing the structural breaks at 2009/05 and 2020/04, and $p = 4$ is selected by likelihood ratio tests. Structural shocks $\boldsymbol{\varepsilon}_t$ are recovered via Cholesky decomposition $\Sigma_u = \mathbf{P} \mathbf{P}'$, so that $\boldsymbol{\varepsilon}_t = \mathbf{P}^{-1} \mathbf{u}_t$ with $\mathbb{E}[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'] = \mathbf{I}_3$. The recursive ordering (Ordering A) restricts MU_t from responding to contemporaneous $IGAE_t$ or FU_t shocks; Ordering B (FU first) is used as a robustness check.

Growth-at-Risk

The quantile estimation, following Adrian et al. (2019) the conditional distribution of IGAE growth is characterized through quantile regressions at horizons $h \in \{1, 3, 6, 12\}$ months. While the impulse responses and local projections obtained in the Structural VAR allow to identify the mean effect of uncertainty shocks on IGAE growth, the quantile regression allows to assess whether uncertainty disproportionately compresses the lower tail of the growth distribution, this under-stress scenario is the most relevant for risk assessment.

For each quantile level $\tau \in (0, 1)$ and forecast horizon $h \in \{1, 3, 6, 12\}$ months, we estimate:

$$Q_\tau(IGAE_{t+h} \mid \mathbf{z}_t) = \alpha_{\tau,h} + \beta_{\tau,h}^{MU} MU_t + \beta_{\tau,h}^{FU} FU_t + \gamma_{\tau,h} IGAE_t, \quad (5)$$

where $g_{t+h}^{(h)}$ denotes the year-on-year growth rate of IGAE at horizon h , g_t is the contemporaneous growth rate (a persistence control), and $\mathbf{z}_t = (MU_t, FU_t, g_t)'$. The quantile regression coefficients are obtained by minimising the asymmetric loss function

$$\hat{\beta}_\tau = \arg \min_{\beta_\tau \in \mathbb{R}^k} \sum_{t=1}^{T-h} \left(\tau \cdot \mathbb{1}_{(y_{t+h} \geq x_t \beta_\tau)} |y_{t+h} - x_t \beta_\tau| + (1 - \tau) \cdot \mathbb{1}_{(y_{t+h} < x_t \beta_\tau)} |y_{t+h} - x_t \beta_\tau| \right) \quad (6)$$

Following Adrian et al. (2019), the direct h -period ahead local projections are estimated via quantile regressions. I estimate the conditional distribution of IGAE growth via quantile

regressions at horizons $h \in \{1, 3, 6, 12\}$ months. I report the 5th percentile of the conditional distribution as the Growth-at-Risk summary statistic. ²

3 Data

3.1 Univariate analysis: IGAE

Figure 1 shows the evolution of Mexican Global Economic Activity Indicator (IGAE acronym from Spanish) for the period January 2004 to December 2024. Instituto Nacional de Estadística y Geografía (INEGI) (2026). The top panel shows the evolution of the variable in levels, where the index is constructed to compare real economic activity in 2018 pesos. The middle panel shows the month-over-month variation, this transformation allows to capture short-term movements and remove the trend of the variable. The bottom panel shows the year-over-year changes, this reflects medium term variation and allows to remove seasonality effects.

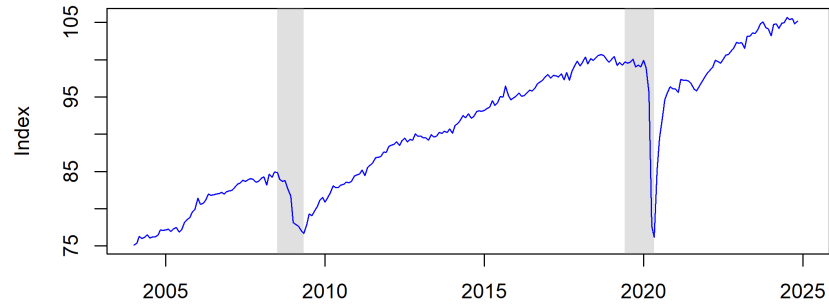
To account for the presence of unit root, I conduct three statistical tests Augmented Dickey-Fuller (ADF), Phillips-Perron (PP) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests; the results from these tests are presented in Table 1. For levels evidence is mixed — ADF/PP reject the unit root at 5% but KPSS also rejects stationarity at 5%. Monthly variation is $I(0)$: ADF and PP reject the unit root, and KPSS does not reject stationarity at any conventional level. Finally, yearly variation is also $I(0)$: ADF and PP reject unit root present, both below the 1% critical value, and KPSS fails to reject stationarity at any conventional level.

We incorporate the results from unit-root tests in two ways. First, this validates using IGAE monthly growth rate in the VAR analysis, since this series is $I(0)$ and satisfies the stationarity requirement for VAR stability. Second, we use IGAE yearly variation in the Growth-at-Risk analysis presented in section 4.5, since it is also $I(0)$ but also allows to smooth over transitory monthly noise and makes a better target for forecasting purposes at different horizons.

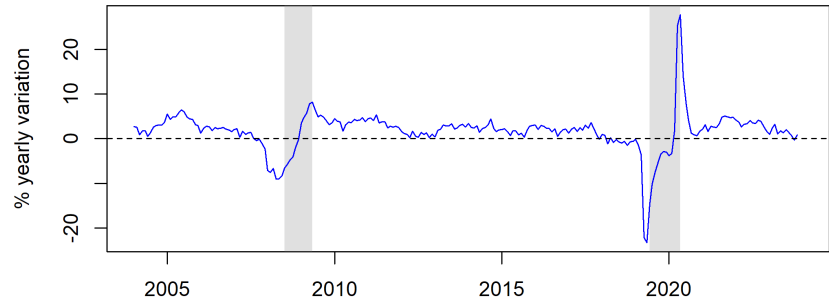
Structural breaks

To account for the presence of both exogenous and endogenous structural breaks I conduct the appropriate statistical tests.

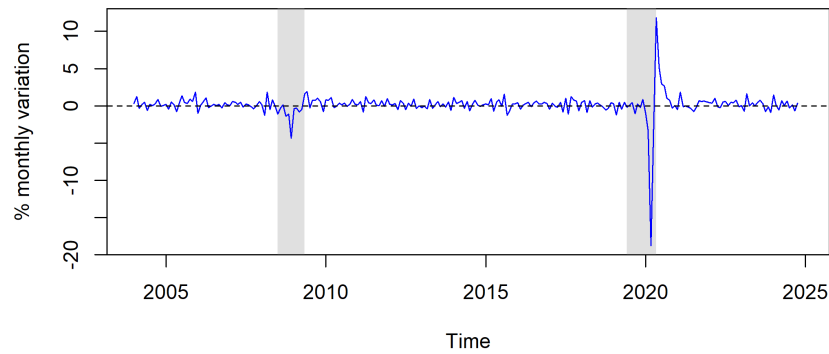
²The parametric skewed- t step of Adrian et al. (2019) is omitted as the quantiles of interest are estimated directly.”



(a) IGAE index (2018 = 100)



(b) Month-on-month growth (%)



(c) Year-on-year growth (%)

Figure 1: INEGI's General Economic Activity Index (IGAE in Spanish), Mexico, 2004–2025. Shaded areas indicate IMEF-dated recession periods (GFC: Jul 2008–May 2009; COVID: Jun 2019–May 2020).

Table 1: IGAE: Unit-root tests

Series	Test	Stat	CV 1%	CV 5%	CV 10%
<i>ADF — H_0: unit root; reject if stat < CV</i>					
Levels	ADF	−4.792	−3.980	−3.420	−3.130
Monthly var.	ADF	−13.030	−3.460	−2.880	−2.570
Yearly var.	ADF	−6.179	−3.460	−2.880	−2.570
<i>PP — H_0: unit root; reject if stat < CV</i>					
Levels	PP	−3.761	−3.998	−3.429	−3.138
Monthly var.	PP	−11.934	−3.458	−2.873	−2.573
Yearly var.	PP	−4.785	−3.459	−2.874	−2.573
<i>KPSS — H_0: stationary; reject if stat > CV</i>					
Levels	KPSS	0.203	0.216	0.176	0.146
Monthly var.	KPSS	0.022	0.739	0.574	0.463
Yearly var.	KPSS	0.080	0.739	0.574	0.463
<i>Notes: ADF - Augmented-Dickey Fuller test, PP - Phillip-Perron test, KPSS - Kwiatkowski-Phillips-Schmidt-Shin test. ADF and PP include trend for levels, drift only for growth rates.</i>					

First, I take care of the dates about the recessions marked by the IMEF’s Mexican Business Cycle Dating Committee (CFCEM, 2026). I apply Chow test at the known recession turning points — this test whether the regression coefficients shift at a *pre-specified* date under the null of parameter stability, the results are presented in Table 2. Structural breaks are present in the *level* of IGAE but not in its monthly growth rate.

Three conclusions arise from this analysis. First, the 2009 and 2020 recessions produced permanent level shifts in economic activity. Second, monthly variation does not reflect this shifts is structurally stable across both episodes ($p > 0.10$). Third, first-differencing simultaneously removes the unit root and the structural breaks.

Table 2: Exogenous Structural Breaks

Series	Date	F	p-val
<i>IGAE levels</i>			
Levels	2009/05	243.86	0.000
Levels	2020/05	115.32	0.000
<i>IGAE monthly variation</i>			
Monthly var.	2009/05	0.271	0.603
Monthly var.	2020/05	1.657	0.199

Notes: Chow parameter stability tests; null hypothesis (H_0) no structural break at the specified candidate date. Candidate breakdown dates identified by the IMEF business cycle recession chronology.

Next, I take care of the presence of endogenous breaks. For which I use Bai-Perron and Zivot-Andrews tests. The Bai-Perron methodology searches for multiple unknown break-points jointly, selecting the number of breaks by BIC and estimating their dates simultaneously, while the Zivot-Andrews test extends the standard ADF by allowing for a single endogenous break in either the intercept, trend, or both. Results are as follows. First, Table 3 shows that Bai-Perron identifies six level shifts in IGAE, concentrated around macro events: 2006/01 (Post-NAFTA adjustment), 2011/05 (Euro-area sovereign crisis), 2014/03 (Oil price decline), 2016/08 (US electoral uncertainty), 2020/02 COVID-19 recession, 2022/03 Post-pandemic inflation surge. Monthly growth rate shows no structural breaks —the BIC selects $m = 0$. Next, Table 4 allows to see that Zivot-Andrews confirms break at 2020/04 ($-5.70 < -5.57$) and rejects the unit root for monthly variation ($-12.80 < -5.34$).

Considering both the results from exogenous and endogenous tests, I added impulse-type dummies in the VAR estimation only at 2009/05 and 2020/04 without distorting the own series dynamics.

Table 3: Bai-Perron (BIC-optimal partition)

	Levels	Monthly var.
Optimal breaks (m)	6	0
BIC	1234	963.7

Notes: Endogenous structural break test results following the Bai-Perron (2003) methodology for IGAE.

Table 4: Zivot-Andrews

Series	Stat	CV 1%	Break
Levels	-5.697	-5.57	2020/02
Monthly var.	-12.797	-5.34	2020/04

Notes: Zivot-Andrews (1992) unit root test, to endogenously determine the single most significant structural break date in the series. Both test statistics reject the unit root null hypothesis at the 1% significance level.

PT decomposition

To perform BN decomposition, I estimate various ARIMA models to identify the best fit, these models are summarized in Table 5. The results show that the AIC gap between ARIMA(2,1,0) and ARIMA(2,1,2) is very small, only 1.2 points. I select upon the BIC

criterion for parsimony. The ARIMA model is $\Delta y_t = \mu + \phi_1 \Delta y_{t-1} + \phi_2 \Delta y_{t-2} + \varepsilon_t$ which has the following BN decomposition closed form:

$$C_t^{\text{BN}} = \frac{(\phi_1 + \phi_2)(\Delta y_t - \mu) + \phi_2(\Delta y_{t-1} - \mu)}{1 - \phi_1 - \phi_2}, \quad (7)$$

with the permanent component $\tau_t^{\text{BN}} = y_t - C_t^{\text{BN}}$. Additional comparison is made with the HP filter using $\lambda = 14400$ given that the frequency of data is monthly. The cyclical components are shown in Figure 2: the cycle obtained under the HP being more volatile than the one with BN. The permanent components are shown in Figure 3: here HP trend component is clearly smoother than the BN, it even to not captures the large decline in the COVID pandemic.

Table 5: ARIMA model selection for IGAE decomposition

Model	N	Loglik	AIC	BIC
ARIMA(2,1,0)	250	679.17	-1352.35	-1341.78
ARIMA(1,1,0)	250	669.50	-1335.00	-1327.96
ARIMA(2,1,2)	250	681.78	-1353.56	-1335.95

Notes: Alternative Autoregressive Integrated Moving Average (ARIMA(p, d, q)) specifications for IGAE. The ARIMA(2,1,0) model is selected for the historical decomposition to prevent overfitting, as dictated by the stricter penalty function of the BIC.

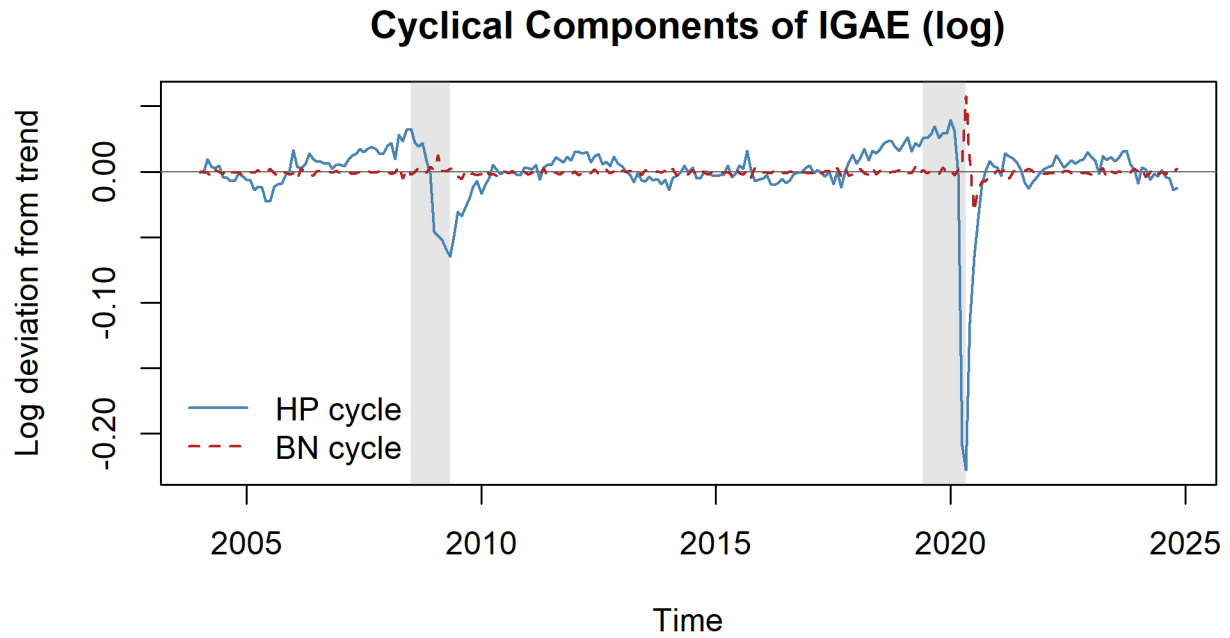


Figure 2: IGAE estimated cyclical components. HP filter using $\lambda = 1600$, BN using ARIMA(2,1,0) specification.

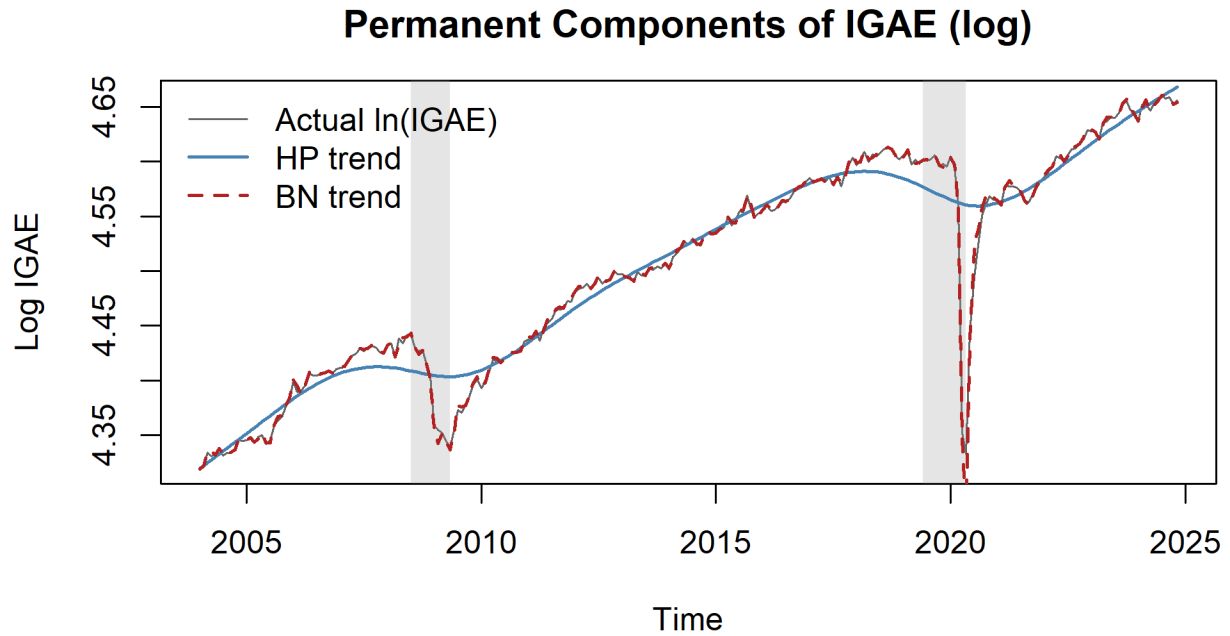


Figure 3: IGAE estimated trend components. HP filter using $\lambda = 1600$, BN using ARIMA(2,1,0) specification.

3.2 Multivariate analysis

The Corona et al. (2022) panel time series consists of 27 variables used to nowcast IGAE on a timely basis, this panel has with the following characteristics: monthly frequency; starting from January 2004, with some available through January 2025; all variables have been obtained in seasonally adjusted form, or if not, they have been adjusted using the X-13ARIMA-SEATS method³. The Table 12 in Appendix shows the description of these variables and the transformation that is considered for the DFM.⁴

For each panel variable, optimal transformation is applied to each series with three possibilities: first-differences, monthly variation, yearly variation such that it maximizes the correlation with IGAE. This transformation is made for two purposes. First, to avoid having to incorporate the target variable into the DFM panel while preserving the economic relationship between the panel variables and economic activity, which is useful for forecasting purposes. Second, this transformation allows to account for the presence of deterministic trends and remove level shifts produced by any structural breaks, thus satisfying the $I(0)$ requirement and to avoid spurious correlations in the DFM.

One departure from Corona et al. (2022) is done, the variables are classified into macro (21), and financial (6) variables. This taxonomy is done after the factors have been obtained, for the purposes of obtaining two uncertainty indexes under the methodology of Jurado et al. (2015), not to perform a hierarchical DFM. The macroeconomic variables included in the analysis are presented in Figure 4, while the financial variables are shown in Figure 5.

³This seasonal adjustment relies on the automatic modeling procedure implemented in the `seas` function from R's `seasonal` package.

⁴I considered including the FFR adjusted for Zero-Lower Bound (Wu & Xia, 2016), the shadow federal funds rate to account; however, this series ends in early 2022. Similarly, the Krippner (2013) (Reserve Bank New Zeland) alternative ends in September 2019 — leaving the COVID tightening cycle (2022–2025) unaddressed by either measure. I therefore exclude the FFR from the panel and note that US monetary policy transmission is partially captured through its effects on the exchange rate (TC), the Mexican short rate (TIE-28), and the S&P 500, all of which are already included.

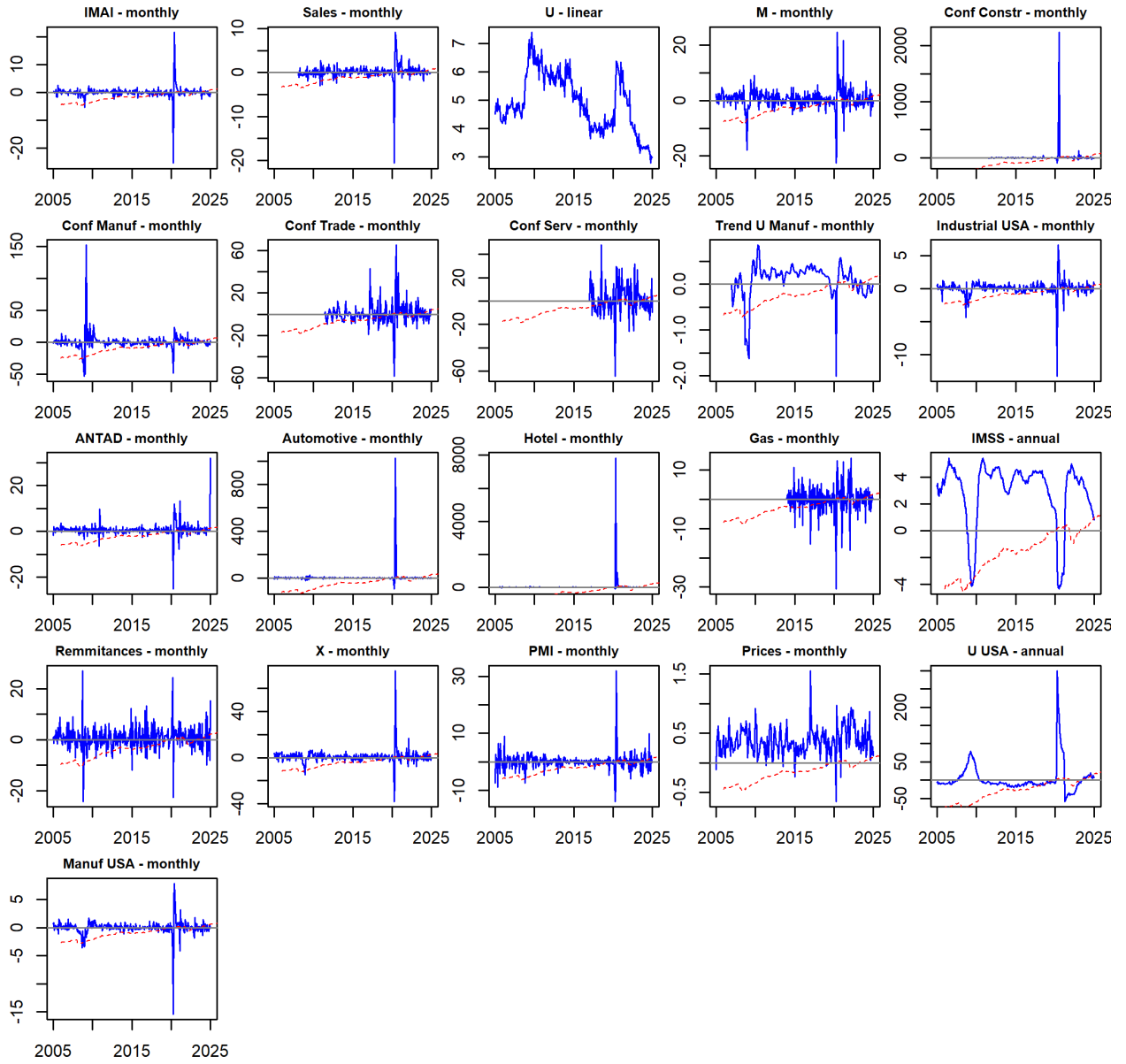


Figure 4: Macroeconomic variables from Corona et al. (2022)

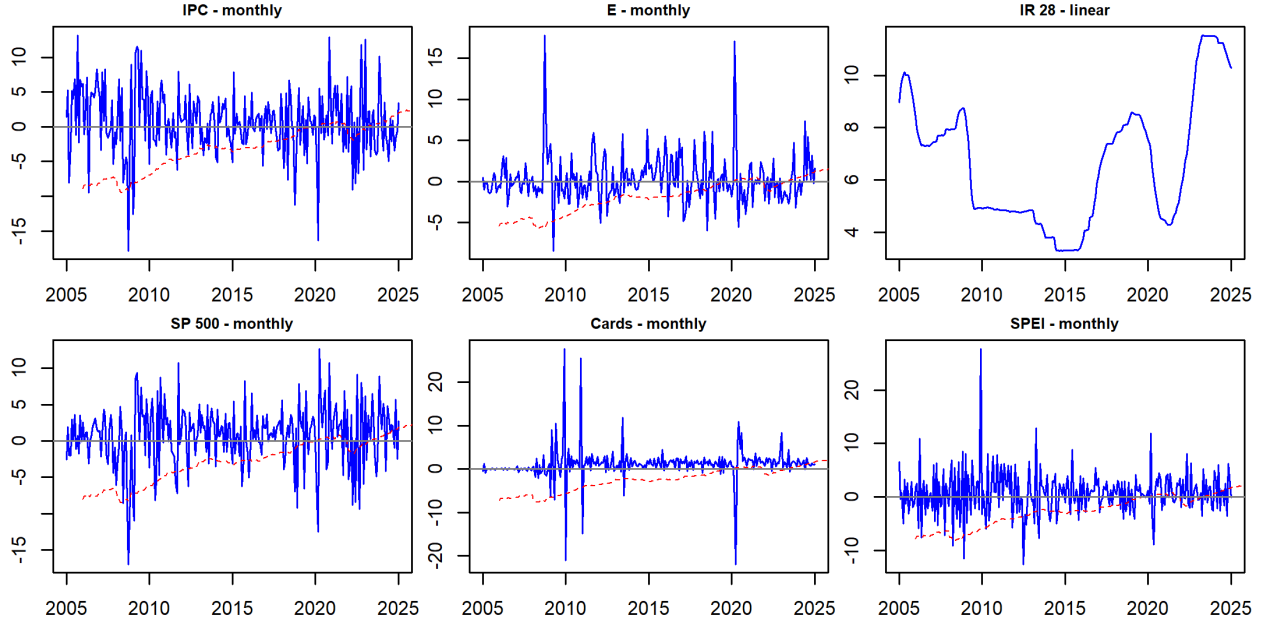


Figure 5: Financial variables from Corona et al. (2022)

4 Results

4.1 Dynamic Factor Model

I first determine the number of factors using the Bai & Ng (2002) methodology, their three criteria minimize a penalized log-variance objective. The first criterion (IC1), includes $(N + T)/NT \cdot \ln(NT/(N + T))$, a penalty for a harmonic mean in N and T dimensions, the second criteria (IC2), includes $(N + T)/NT \cdot \ln(\min(N, T))$, which imposes a higher penalty when either dimension is small, while the third criteria (IC3), $\ln(\min(N, T))/\min(N, T)$, grows relatively slowly with the panel size and therefore tends to over-select factors. The results are shown in Table 6: IC1 and IC2 select $r = 7$, whilst IC3 suggests $r = 8$. I decide to take 7, by majority of criteria. The seven factors explain 66.9% of total panel variance, while the remaining 33.1% is idiosyncratic variance, and is to be used for the GARCH modeling under the JLN methodology.

In addition, Figure 6 show the factor loadings that allow for interpretation of the results. We see that the first factor, F1, is mostly related to activities in the transformation sectors in the economy both for the Mexican and the US economy: industrial production, manufacturing, automobile production, exports and imports, we can even refer to it as the Mexican business cycle. The second factor, F2, is mainly related to unemployment both in Mexico

Table 6: Determining the number of factors Bai & Ng (2002) criteria

r	IC1	IC2	IC3
0	-0.004	-0.004	-0.004
1	-0.191	-0.186	-0.200
2	-0.200	-0.192	-0.219
3	-0.205	-0.192	-0.233
4	-0.208	-0.190	-0.245
5	-0.215	-0.193	-0.262
6	-0.218	-0.192	-0.274
7	-0.233	-0.202	-0.298
8	-0.228	-0.193	-0.302

Notes: Information criteria to determine the optimal number of common latent factors (r) to extract from the macroeconomic panel data used for constructing the financial indices. Bold values indicate the minimum value achieved by each respective criterion, identifying the optimal factor allocation.

and in the US, and stock markets, also in both countries, also it is negatively correlated to the level of formal employment registration in Mexico, IMSS; this can be called as labor market stress + equity markets. The third factor, F3, relates the most to exchange rate, US unemployment and remittances, while at the same time negatively related to stock markets, employment in the manufacturing industry and the formal employment registry, IMSS; this factor captures the effect of peso depreciation and remittances. The fourth factor, F4, considers mainly the statistical relationship to the interest rate and negatively related to the unemployment rate; clearly reflects monetary policy and interest rate channel. The fifth one, F5, is associated with lodging services, *Hotel*, but negatively related to automobile production, manufacturing, exports and imports; this factor mostly reveals the relation between domestic sector versus external manufacturing factor. The sixth factor, F6, is negatively related to banking information, cards and SPEI, and in lesser amount to manufacturing confidence; this factor accounts for credit and financial conditions. Finally, the seventh factor, F7, is related to business and consumer sentiment, with confidence indexes for consumers, commerce, services, and manufacturing all loading positively.

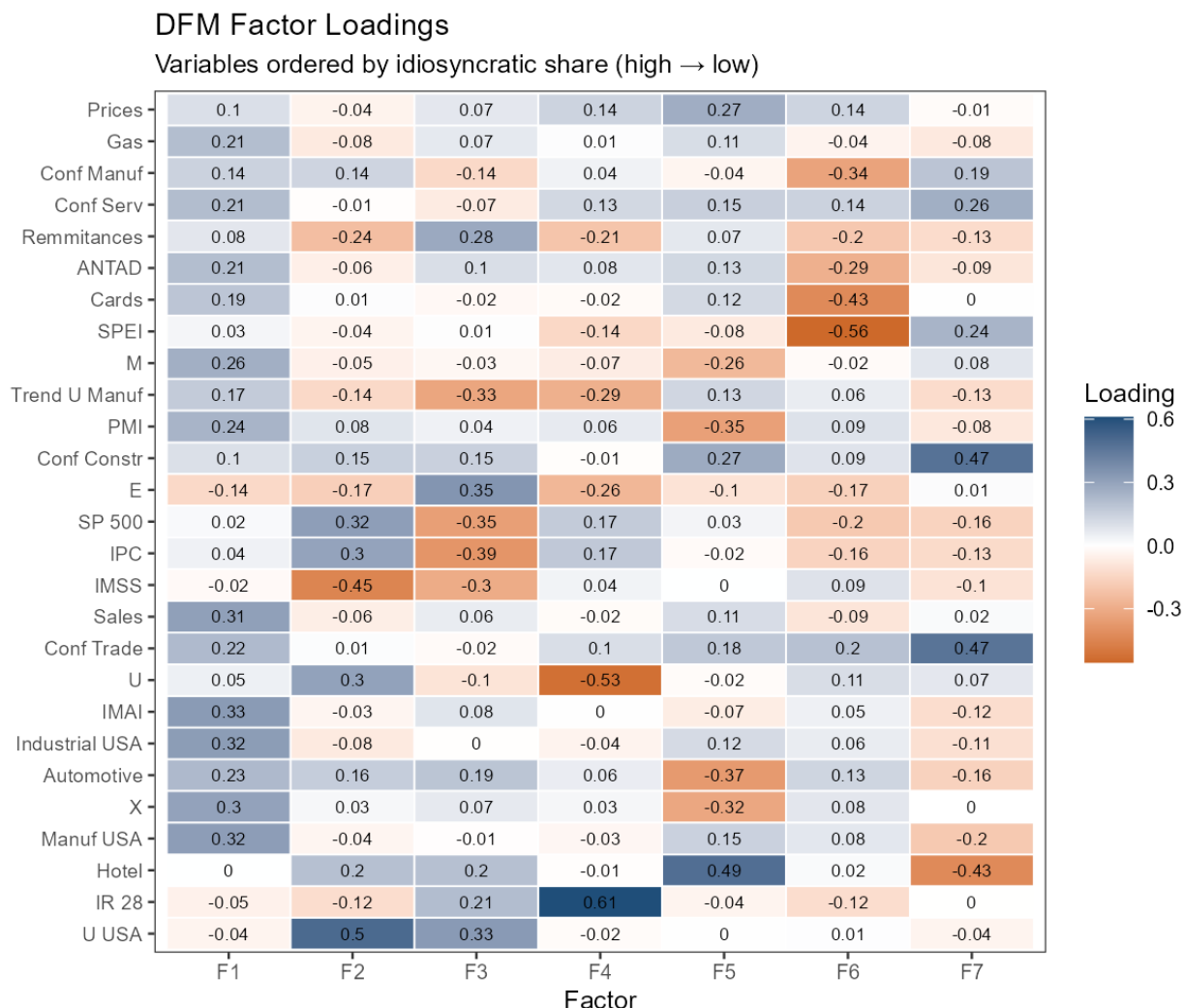


Figure 6: DFM factor loadings

Given these interpretation to the seven estimated factors, in Figure 7 I show the evolution of them. Note that three of them —F1 - business cycle, F5 - domestic vs external sector, F7 - business and consumer sentiment— vary alongside the dynamic of IGAE monthly variations shown in the bottom of Figure 1. The factors F2 and F3 clearly capture the large swings during the recessions commented before, the Great Recession and COVID-19 pandemic, consistent with the simultaneous rise in unemployment and currency pressure during both episodes. The factor F4, Notably, F6 - credit and financial conditions, is the only one that do not reflects the large falls during recessions, and captures persistent background volatility throughout the sample given that it does not capture the large tail events.

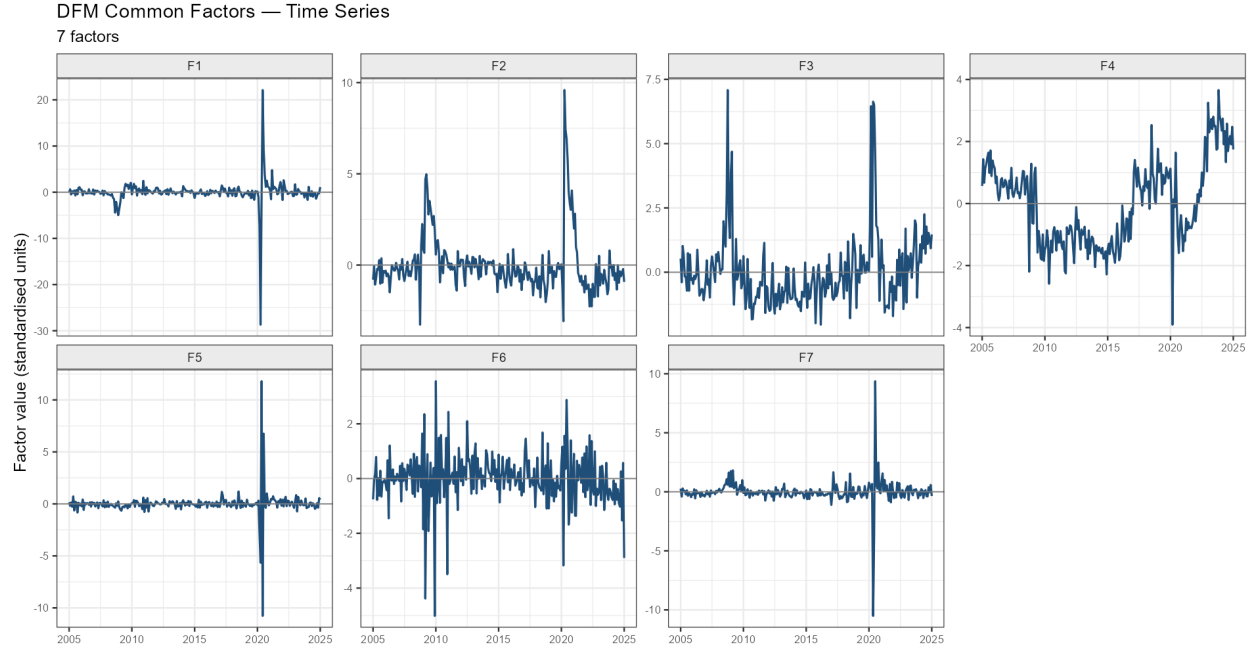


Figure 7: Factors extracted for the 27 variables panel for the Mexican economy

4.2 Macroeconomic and Financial Uncertainty

Once the DFM have been estimated, the idiosyncratic component for variable i at time t is obtained as a residual, which are later aggregated into uncertainty measures for the macroeconomic conditions and the financial conditions of the Mexican economy. Figure 8 shows these estimations along with the monthly variations of IGAE.

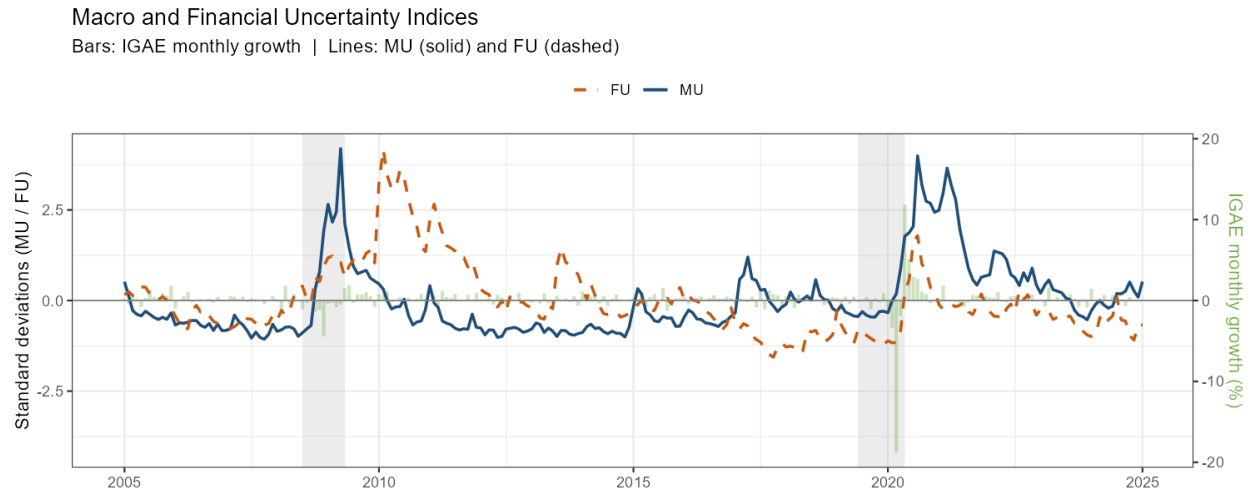


Figure 8: Macroeconomic and Financial Uncertainty indexes for the Mexican economy

By construction, the uncertainty indicators capture the unforeseeable component of the

macroeconomy and financial conditions. Both indices are standardized, thus values below zero indicate uncertainty is below its historical averages; values above indicate high uncertainty episodes.

MU indicator remains in general near or below zero throughout most sample, consistent with Mexico's relatively stable macro conditions in the period inter-crisis (2010-2018). Some exceptions are the spikes around the beginning of the GFC and modest upward react to political turmoil events as the 2012 and 2018 Mexican elections and the 2016 US elections. The post COVID pandemic recession, shows that this indicator has been on positive terrain, indicating a period of relative higher macroeconomic uncertainty.

In the case of FU, we see that this indicator reacted earlier in the GFC, but its spike was not during the main period of recession, but during the recovery part around 2009-2010. Similarly, this indicator increased around 2013-2014 on its own, while the MU remained on its regular level below zero. In addition, after the COVID-19 episode, the FU indicator, increased but returned very quickly and have remained below the zero afterwards. This divergence suggests that financial markets priced in and resolved the COVID uncertainty relatively fast, while real macroeconomic conditions remained structurally less predictable.

As a benchmark comparison, in Figure 9 I compare the MU and FU indices to the one available for the Mexican economy under the Baker et al. (2016) methodology. The EPU is mostly related to the MU than to the FU; note, however, that the EPU is text-based and reacts to news flow, while MU measures realized unpredictability in macro data, which took longer to subside. In particular, both spike sharply and simultaneously at the beginning of GFC (2008); similarly, both spike and both remain elevated in the post-pandemic period through 2022–2024; during the period 2017–2018, both show a moderate rise, consistent with NAFTA renegotiation uncertainty and the 2018 Mexican election; finally, in recent times both turning upward, consistent with US tariff and trade policy uncertainty.

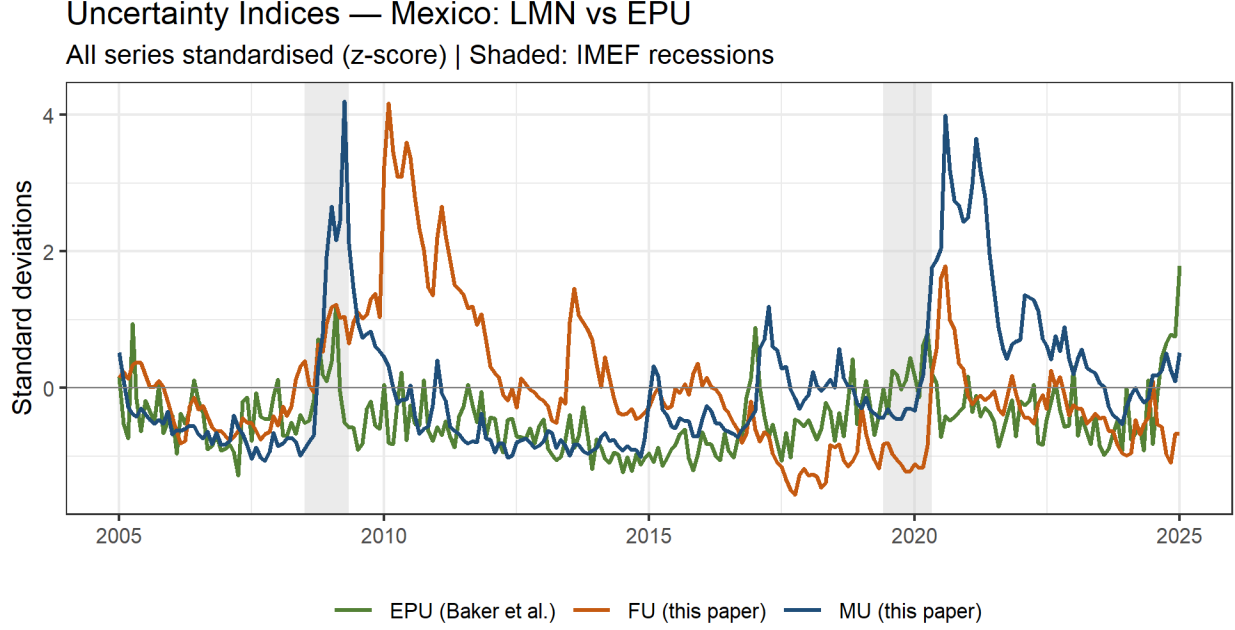


Figure 9: Comparison with the Mexican EPU under Baker et al. (2016) methodology.

4.3 VAR estimation

The VAR estimation is performed in accordance to Ludvigson et al. (2021) framework in which the Cholesky ordering of variables is $MU \rightarrow IGAE \rightarrow FU$ (A: MU first). Similarly, an alternative VAR ordering is tested, in which the FU precedes IGAE and MU, respectively (B: FU first). The baseline analysis is conducted with specification A, and then the model is estimated with B specification for comparison.

To proceed with VAR estimation, I first added the impulse-type dummies from the exogenous and endogenous tests in section 3.1. Then, to select the appropriate number of lags I compare three different information criteria: Akaike (AIC), Hannan and Quinn (HQ) and Schwarz (SC). The results from these tests are presented in Table 7: AIC selects $p = 5$, HQ selects $p = 2$, and SC selects $p = 1$. Note that, with $p = 1$ and $K = 3$ variables the VAR has $K \times (K + 1) = 12$ free parameters; thus, I use as the baseline for parsimony I use $p = 1$.

Table 7: Lag selection with different information criteria

p	AIC	HQ	SC
1	-3.54	-3.44	- 3.28
2	-3.65	- 3.48	-3.25
3	-3.61	-3.40	-3.08
4	-3.71	-3.44	-3.04
5	- 3.76	-3.43	-2.95
6	-3.73	-3.35	-2.79

Notes: Bold indicates the selected lag for each criterion. VAR model: $\mathbf{X}_t = (MU_t, IGAE_t, FU_t)'$, $p_{\max} = 6$. Impulse dummies for 2009/05 and 2020/04 included as exogenous regressors. Baseline uses $p = 1$ (SC); robustness at $p = 5$ (AIC).

In addition, I perform likelihood ratio tests to determine whether additional lags improve the model. The LR test compares a restricted VAR(p_r) against an unrestricted VAR(p_u), with null hypothesis that the additional coefficients matrices are jointly zero — the restricted model fits as well as the unrestricted one. The results presented in 8 show that adding lag 2 over lag 1 is statistically significant (p), but lag 3 does not add with respect to lag 2. Lag 4 then re-enters significantly (LR = 41.16 over $p = 3$, $p < 0.001$), and the joint test $p = 1$ vs $p = 4$ strongly rejects parsimony (LR = 92.18, $p < 0.001$). Thus, although SC selects $p = 1$, the joint rejection of $p = 1$ against $p = 4$ (LR = 92.18, $p < 0.001$) and the richer impulse response dynamics at $p = 4$ shown in next section lead me to adopt $p = 4$ as the baseline specification; $p = 1$ is retained as a parsimony robustness check.

Table 8: Likelihood ratio tests for VAR lag order

H_0	H_1	LR stat	df	p -value
$p = 1$	$p = 2$	41.86	9	0.000***
$p = 2$	$p = 3$	9.75	9	0.371
$p = 3$	$p = 4$	41.16	9	0.000***
$p = 3$	$p = 5$	69.06	18	0.000***
$p = 1$	$p = 4$	92.18	27	0.000***

Notes: LR = $T(\ln|\hat{\Sigma}_r| - \ln|\hat{\Sigma}_u|) \sim \chi^2(K^2\Delta p)$, where $K = 3$. All models estimated on a common sample (rows $p_{\max} + 1$ to T) with dummies for 2009/05 and 2020/04 included as exogenous regressors. *** $p < 0.01$.

Results for VAR estimation are shown in Figure 10 at $p = 4$. Top panel, 10a, shows impulse response functions for both specifications; irrespective of ordering, the two plots show that neither uncertainty channel produces a statistically significant response of IGAE growth at any horizon: zero lies within the 90% bootstrap bands throughout both panels. Under ordering A, a one-standard-deviation MU shock reveals a small negative impact on

IGAE that recovers around months 4–5, consistent with a wait-and-see mechanism, as in the literature of the EPU; in addition, the FU shock produces a positive impact response under both orderings, which is counterintuitive. Under ordering B, responses are similar in direction but with different magnitude. MU responses are stable across orderings, while FU responses are larger in Panel B, indicating that that FU’s influence on IGAE operates primarily within the same month and is suppressed by the Cholesky restriction when FU is ordered last.

The bottom panel, 10b, shows the forecast variance error decomposition for both specifications. In any case, IGAE own dynamics account for more than 98% of the FEVD, leaving less than 2% to the uncertainty indicators. MU contributes approximately 0.8% at the 24-month horizon and is insensitive to ordering. FU is ordering-sensitive: its share is approximately 0.65% under Ordering A but rises to approximately 1.35% under Ordering B.

4.4 Local Projection

As a robustness check, I estimate IRF under the approach of Jordà (2005). This methodology rather than recovering IRFs from the VAR’s recursive structure, uses the orthogonal shocks of the VAR model to compute the response at each horizon, by projecting the outcome h periods ahead on the structural shock today.

In this case the estimated equation at each horizon $h = 0, 1, \dots, 24$ is:

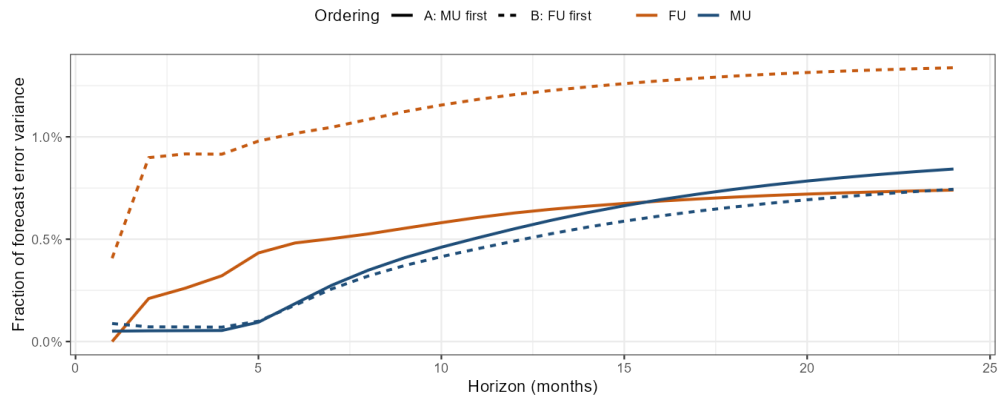
$$IGAE_{t+h} = \alpha_h + \beta_h^{MU} \cdot \varepsilon_t^{MU} + \beta_h^{FU} \cdot \varepsilon_t^{FU} + \sum_{j=1}^p \gamma'_{h,j} \mathbf{X}_{t-j} + e_{t+h}$$

where $\mathbf{X}_{t-j} = (MU_{t-j}, IGAE_{t-j}, FU_{t-j})'$ are $p = 4$ lags of the VAR variables included as controls. As seen in Figure 11, the results do not vary significantly from the obtained above: point estimates are close to zero at all horizons and statistically insignificant throughout.



(a) Impulse Response Function

FEVD: Share of IGAE Variance Explained by MU and FU



(b) Forecast Error Variance Decomposition

Figure 10: VAR estimation. Variables included $MU \rightarrow IGAE \rightarrow FU$

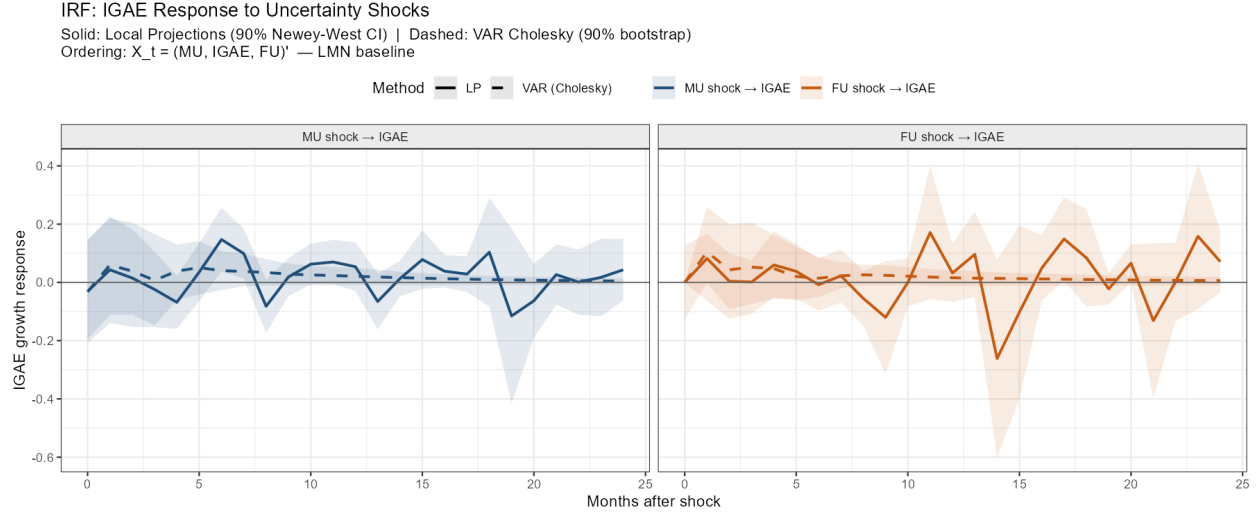


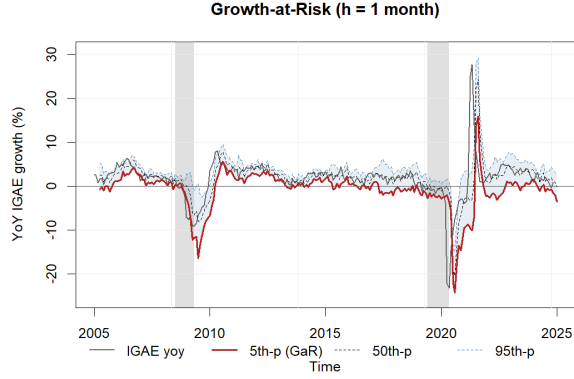
Figure 11: IRF estimated with Local Projections methodology

4.5 Growth-at-Risk

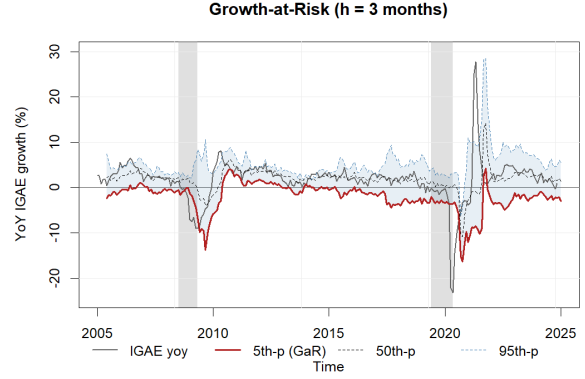
Once I have constructed the uncertainty indicators, MU and FU, I can incorporate them into the Adrian et al. (2019) framework to assess their relevance into the forecast of IGAE growth conditional on current conditions and uncertainty.

The results of these quantile regressions are presented in Figure 12 for $h \in \{1, 3, 6, 12\}$ months. Three features are consistent across all horizons. First, the GaR estimate (5th percentile, red) lies persistently below zero even in periods of no stress, this reveals that downside risk is always present, not only during recessions. Second, both the GFC and the COVID-19 recession are captured by sharp deteriorations in the lower tail, confirming that MU and FU have predictive content for extreme negative growth outcomes. Third, the range between the 5th and 95th percentile widens substantially with the horizon; at $h = 1$ the uncertainty conditions contain most of the near-term distributional information, and expands at $h = 12$, reflecting genuine uncertainty about growth a year ahead.

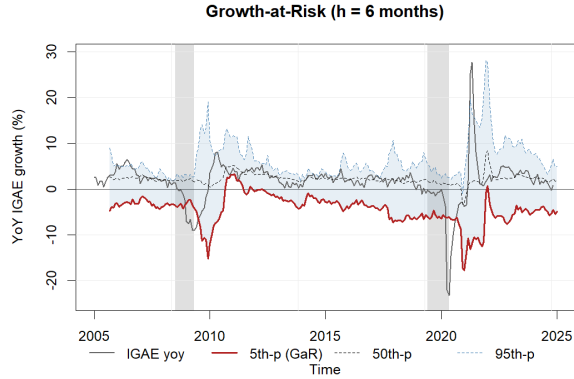
In addition, the horizon dimension also reveals important asymmetries. At $h = 1$, the GaR is near zero in tranquil periods and collapses to approximately -24% during COVID. At $h = 12$, the GaR is persistently around -7 to -9% throughout the sample, including in expansion phases, and the 95th percentile expands to $+25\%$ during the post-GFC recovery — indicating that a year-ahead outlook carries both large downside and large upside risk.



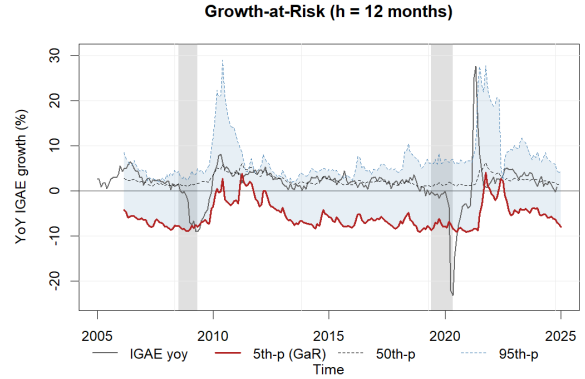
(a) $h = 1$ month



(b) $h = 3$ months



(c) $h = 6$ months



(d) $h = 12$ months

Figure 12: Growth-at-Risk: conditional distribution of year-on-year IGAE growth at different horizons. Shaded band: 10th–90th percentile fan. Red solid line: 10th percentile (downside risk). Black dashed: median (50th percentile). Blue dashed: 90th percentile. Grey solid: realized YoY IGAE growth. Shaded areas: IMEF recession dates. Conditioning variables: MU, FU, lagged YoY growth.

4.6 Cointegration analysis

In this section of the paper, I test whether there is a long-run relationship between economic activity —measured by the monthly proxy of GDP—and employment —using registry for formal employment—in the Mexican economy, taking into account the uncertainty indices constructed in the previous sections.

First, I establish the integration order of all four series using augmented Dickey-Fuller tests complemented by Zivot-Andrews (1992) tests that allow for one endogenous structural break. Then, I test for block-exogeneity Granger causality. Cointegration is examined by comparing the residual-based Engle-Granger test with the full-information maximum likelihood Johansen (1991) approach, which determines the cointegrating rank jointly. Given evidence of cointegration, I estimate a vector error-correction model (VECM) and analyse the cointegrating vector and the loading matrix which governs the speed of adjustment of each variable toward the long-run equilibrium.

Unit-root tests

In Table 9 shows the results for unit root tests to be included in the cointegration analysis. IGAE and IMSS are $I(1)$, the first one based on the analysis done before for structural break, the second from the test herein with trend. In the case of uncertainty indices MU it is $I(0)$, while FU is $I(1)$ these two variables are tested with drift.

Table 9: ADF unit-root tests.

Series	k	τ (levels)	k	τ (Δ)	Order
log(IGAE)	1	-4.791	1	-12.616	$I(1)^\dagger$
log(IMSS)	1	-2.207	1	-12.842	$I(1)$
MU	1	-2.914	1	-10.240	$I(1)^\ddagger$
FU	1	-2.735	1	-11.212	$I(1)^\ddagger$

Critical values (5%): $\tau_3 = -3.45$, $\tau_2 = -2.88$
(MacKinnon 1996).

Block-exogeneity Granger causality

The results from block exogeneity are detailed in Table 10. First, we test whether uncertainty —MU, FU— causes real economic activity —IGAE, IMSS—. The result of this test reject the null hypothesis, p-value= 0.0001948, that is, uncertainty measures possesses information to predict economic activity and formal employment. Next, we test the null hypothesis that

IGAE does not cause $\{IMSS\}$, and viceversa; we reject the null in both cases, $p - val = \{2.282e - 12, 1.802e - 11\}$.

Table 10: Block-exogeneity Granger causality tests.

H_0	Test	Statistic	df_1	df_2	p -value
$\{MU, FU\} \not\rightarrow \{\log IGAE, \log IMSS\}$	F	2.5681	20.00	844.00	0.0002***
$\log IGAE \not\rightarrow \{\log IMSS, MU, FU\}$	F	6.1384	15.00	844.00	0.0000***
$\log IMSS \not\rightarrow \{\log IGAE, MU, FU\}$	F	5.7835	15.00	844.00	0.0000***

VAR(5) in levels; exogenous: D_{2009}, D_{2020} . Significance: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Testing cointegration

The cointegration relationship estimated is

$$\log(IGAE_t) = -4.1913 + 0.5224 \cdot \log(IMSS_t) + -0.0169 \cdot MU_t + -0.0030 \cdot FU_t$$

The ADF test on the residuals of this estimation gives a test-statistic of -7.4908 with 1 lag, which is smaller than the MacKinnon (1991) critical values ($1\% = -4.64, 5\% = -4.10, 10\% = -3.81$). We fail to reject the null hypothesis of no cointegration—at least one CI vector found—at the 5% level of significance.

The Johansen procedure rejects the null hypothesis of zero and less than one cointegration vectors, but fails to reject the null of less than or equal to two cointegration vectors. The lambda trace ($\lambda_{trace} = 14.96$) and lambda max $\lambda_{max} = 8.660$ are both smaller than their critical values 19.96 and 15.67, respectively.

Table 11: Johansen cointegration tests.

H_0	λ_{trace}	CV 5%	λ_{max}	CV 5%
$r = 0$	201.074	53.12	114.679	28.14
$r \leq 1$	86.394	34.91	71.455	22.00
$r \leq 2$	14.939	19.96	8.660	15.67
$r \leq 3$	6.279	9.24	6.279	9.24

VAR lag $K = 4$, Exogenous: D_{2009}, D_{2020} .

VECM

The VECM analysis allows to single out both the long-term relationships and the short-run velocity of adjustment of the variables in the system.

The normalized CI vector captures the long-run relationship is as shown below. Once rearranged, the coefficients can be interpreted as follows: there is a positive long-run relationship between IMSS employment registry and IGAE, which is less than unity. This smaller than one value may be attributable to the fact that this variable captures only the formal employment in the economy, which can represents around 50% of total employment. During economic turmoils informal employment works as a buffer for formal employment, workers migrate from formal to informal employment. The small and negative coefficients associated to both uncertainty indices confirm the results found in the SVAR section —negative but modest reaction of IGAE to uncertainty.

$$\beta'Y_t = \begin{pmatrix} 1.0000 & -0.5223 & 0.0134 & 0.0028 & 4.1894 \end{pmatrix} \begin{pmatrix} \log(\text{IGAE}_t) \\ \log(\text{IMSS}_t) \\ \text{MU}_t \\ \text{FU}_t \\ \text{const} \end{pmatrix} = 0$$

Finally, the estimated vector of speed of adjustment is shown next. Given a deviation in the system of one unit from the long-run equilibrium, IGAE adjust toward equilibrium at a rate of -0.3 pp in the next period. On the contrary, IMSS adjustment coefficient is very small and positive, 0.03 , this reflects the rigidity of formal labor markets in Mexico. Notably, the short-run adjustment to uncertainty are large negative; the large coefficient for MU -2.71 captures the endogeneity of uncertainty to the business cycle, similarly, FU shows a large coefficient -2.5 . These last two results indicate that macro uncertainty is somewhat more responsive to real activity disequilibria than financial uncertainty.

$$\hat{\alpha} = \begin{pmatrix} \Delta(\log(\text{IGAE}_t)) \\ \Delta(\log(\text{IMSS}_t)) \\ \Delta(\text{MU}_t) \\ \Delta(\text{FU}_t) \end{pmatrix} = \begin{pmatrix} -0.3003251 \\ 0.02855754 \\ -2.713334 \\ -2.051611 \end{pmatrix}$$

5 Conclusions

In this paper, I extended the methodology of Corona et al. (2022) to estimate uncertainty measures from the idiosyncratic components in the DFM used to nowcast short-term economic activity indicator in Mexico. The methodology of JLN/LMN allowed me to build uncertainty indices that resemble the one obtained by Baker et al. (2016) for the Mexican economy.

This indices have three main features. The two indices, macro uncertainty and financial

uncertainty capture separable dimensions of aggregate risk. MU co-moves closely with text-based policy uncertainty measures, while FU captures idiosyncratic financial stress episodes

The creation of such indices is employed in the paper for analyzing short-run dynamics identification, through SVAR IRF and Local Projections: both MU and FU small generate negative shocks that peak at two to four months and dissipate within a year. The forecast error variance decomposition finds modest share of IGAE fluctuations to uncertainty shocks. The Johansen FIML cointegration tests reveal a stable long-run relationship linking economic activity, formal employment and both uncertainty indices. Finally, Growth-at-Risk analysis shows that elevated uncertainty disproportionately compresses the lower tail of the IGAE growth distribution across all horizons.

Further directions in this research include adding the Federal Funds Rate for the US adjusted for zero lower bound, to control for US monetary policy spillovers transmitted to Mexico through capital flows. Also, completing the two-step Adrian et al. (2019) procedure by fitting a skewed- t distribution to the quantile fan at each horizon.

References

- Adrian, T., Boyarchenko, N., & Giannone, D. (2019). Vulnerable growth. *American Economic Review*, 109(4), 1263–1289.
- Bai, J., & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4), 1593–1636.
- CFCEM. (2026). *Ciclos economicos* (Tech. Rep.). Comité de Fechado de Ciclos de la Economía de México (CFCEM-IMEF). Retrieved from <https://comitefechadocicloseconomicos.mx/fechado-de-ciclos/> (Accessed: May 8, 2026)
- Corona, F., González-Farías, G., & López-Pérez, J. (2022). Timely estimates of the monthly mexican economic activity. *Journal of Official Statistics*, 38(3), 733–765.
- Doz, C., Giannone, D., & Reichlin, L. (2012). A quasi-maximum likelihood approach for large, approximate dynamic factor models. *Review of economics and statistics*, 94(4), 1014–1024.
- Elizondo, R. (2019). Estimaciones del pib mensual en méxico basadas en el igae. *Estudios Económicos (México, DF)*, 34(2), 197–241.

- Instituto Nacional de Estadística y Geografía (INEGI). (2026). *Producto Interno Bruto trimestral, base 2018: Serie desestacionalizada a precios de 2018, total*. Retrieved from https://inegi.org.mx/app/indicadores/?tm=3#D736181_600609 (Series calculadas por métodos econométricos a partir de las cifras generadas por el Sistema de Cuentas Nacionales de México. Periodicidad trimestral, millones de pesos a precios de 2018. Última actualización: 23 de febrero de 2026. Accessed: April 10, 2026)
- Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American economic review*, 95(1), 161–182.
- Jurado, K., Ludvigson, S. C., & Ng, S. (2015). Measuring uncertainty. *American Economic Review*, 105(3), 1177–1216.
- Krippner, L. (2013). Measuring the stance of monetary policy in zero lower bound environments. *Economics Letters*, 118(1), 135–138.
- Ludvigson, S. C., Ma, S., & Ng, S. (2021). Uncertainty and business cycles: exogenous impulse or endogenous response? *American Economic Journal: Macroeconomics*, 13(4), 369–410.
- Wu, J. C., & Xia, F. D. (2016). Measuring the macroeconomic impact of monetary policy at the zero lower bound. *Journal of Money, Credit and Banking*, 48(2-3), 253–291.

Table 12: Variables to the DFM

Variable	Description	Trans.
<i>Macro variables</i>		
IMAI	Industrial production	monthly
Sales	Retail sales	monthly
U	Unemployment	monthly
M	Imports	monthly
Conf Constr	Construction confidence	monthly
Conf Manuf	Manufacturing confidence	monthly
Conf Trade	Trade confidence	monthly
Conf Serv	Services confidence	monthly
Trend L Manuf	Manuf. employment trend	monthly
Industrial USA	U.S. industrial production	monthly
ANTAD	ANTAD retail sales	monthly
Automotive	Vehicle production	monthly
Hotel	Hotel occupancy	linear
Gas	Gasoline sales	monthly
IMSS	IMSS formal employment	monthly
Remittances	Remittances inflows	monthly
X	Exports	monthly
PMI	Manufacturing orders	monthly
INPC	Prices	monthly
U USA	U.S. unemployment rate	monthly
Manuf USA	U.S. manufacturing activity	monthly
<i>Financial variables</i>		
IPC	Stock exchange index	annual
E	Exchange rate	monthly
IR 28	Short-term interest rate	annual
SP 500	S&P 500 Index	annual
Cards	Credit/debit card transactions	monthly
SPEI	Interbank electronic transfers	monthly